

Signal Detection
PSY 310: Lab in Psychology
Lab Report

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Introduction

Signal Detection theory is a theoretical form of measurement to understand how systems and people discriminate between signal and noise under conditions of uncertainty. Noise signals are distractors and responses are classified into binary categories (Vickers, 2014). The core concepts are 1) signal which is stimulus to be detected 2) noise which is background interference 3) sensitivity (d') which is ability to tell a signal apart and 4) decision criterion (c or bias) which is willingness to yes the signal is present based on expectations, fatigue or stakes. There can be four outcomes 1) saying yes when signal is present 2) no when signal is present 3) yes when signal is absent and 4) saying no when signal is absent.

Method

Participant

The study had one subject, who was a female, 43 years old. Their informed consent was taken prior to the experiment and confidentiality was assured.

Materials

The study was conducted using PsychoPy v2026.2.3. It was conducted on a laptop with a monitor of dimensions 12.5", with a resolution of 1920x1080 pixels and a refresh rate of 60Hz.

Procedure

Participants were given instructions explaining the task and required response. The participants viewed the stimulus, triangle, for one second and responded using the up arrow key or down arrow key. The direction of tilt was randomly determined with equal probability. The experiment used a sinusoidal grating presented for 0.3 seconds either oriented vertically, so press up key, or tilted slightly to the left or right, press down key, by a randomly selected angle between 1 and 5 degrees. There were 100 trials in a random order, responses were recorded by PsychoPy, along with accuracy of response. The orientation was also recorded and analysed.

Results

Hits	17
Misses	38
False Alarms	10

Correct Rejections	34
Hit Rate	$17/55 = 0.309$
False Alarm Rate	$10/44 = 0.227$

Using standard signal detection theory:

$$d' = Z(H) - Z(FA)$$

$$d' = Z(0.309) - Z(0.227)$$

$$d' = 0.25$$

Criterion:

$$c = -0.5[Z(H) - Z(FA)]$$

$$c = -0.62$$

Discussion

The results show that the participant has relatively low sensitivity in differentiating between upright and tilted gratings. The $d' = 0.25$ suggests the capability to differentiate between signal present and absent, however trials was only slightly above chance level. This can be attributed to the tilted stimulus angles range being short. There were 17 hits and 10 false alarms, suggesting some stimuli was correctly responded to but considerable responses were incorrect. The c value of -0.62 shows a response bias towards reporting that signal was present as participants were relatively more willing to respond that a stimulus was tilted showing that the performance in the experiment was influenced by both response bias and perceptual sensitivity. In conclusion, the experiment shows the importance of signal detection theory in separating an individual's ability to differentiate between stimuli from their tendency to respond to adopt a particular response criterion.

References

Vickers, D. (1979). *Decision processes in visual perception*. Academic Press.